

TATA RANGESURE

Turning real-world range into a fleet advantage

AI-Native Transformation Strategy | Commercial Vehicles

DTAI Capstone 2026 · IIM Lucknow



26°C

Light Rain

80%



80%

Battery

244 km

Est. Full Range



MUMBAI → PUNE

MAX ELEVATION
622 m



BATTERY

80%

244 km

Est. Full Range



ROAD

NH 48



SPEED

60 km/h

Avg.



TELEMATICS

Connected



ENERGY

0.82 kWh/km

Make real-world range predictable before asking fleets to scale electric

1 THE PROBLEM



Certified range $\neq$ route reality



Payload, terrain and traffic change every trip



One unplanned charge can derail the delivery window

» Fleets hesitate because tomorrow's trip still feels **uncertain**.

2 THE ANSWER

TATA RANGESURE

Predict the trip. Plan the charge. Protect the delivery.



3 THE PAYOFF



Lower ₹/km



Fewer unplanned stops



Higher fleet uptime



More e-truck conversions

1 Range confidence

2 Predictable trip economics

3 Cost-per-km contracts



BOARD ASK



Approve a 12-month pilot with selected fleet customers; scale only after **range accuracy** and **savings** are proven.



AI predicts energy use; the driver keeps final control.

Tata already owns the vehicle and data layer; the missing piece is **reliable trip-level guidance**

MARKET & VEHICLE BASE



36.8%

Domestic CV market share



3,400+

eCV orders across segments



ACE EV • INTRA EV • ULTRA E • PRIMA E

WHAT TATA HAS TODAY

1



ELECTRIC VEHICLES

Battery, motor, regenerative braking

2



FLEET EDGE

Location, vehicle health, driving behaviour

3



LIVE VISIBILITY

Track what is happening

?



MISSING: TRIP DECISION

Data exists, but drivers still combine range, load, weather and charging manually.

WHAT RANGESURE ADDS

TATA RANGESURE

Know what to do next



Can this trip be completed?



Which route uses least energy?



Where should the truck charge?



What arrival battery should we expect?



600,000+

vehicles connected to Fleet Edge



REAL-TIME

vehicle and driver insights



READY TO SCALE

No new hardware required for the pilot



Fleet Edge explains the vehicle. **RangeSure** turns that data into a **trip decision**.

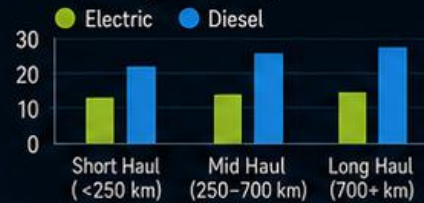
Electric-truck economics are improving; operational uncertainty is now the **adoption bottleneck**

ECONOMICS

₹7–18/km

Potential operating-cost advantage across truck applications

Total operating cost (₹/km)



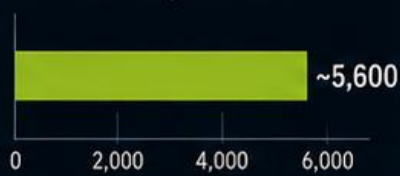
↓ Lower running cost makes the business case possible.

POLICY

₹9.6 lakh

Maximum incentive per eligible e-truck

PM E-DRIVE target (e-trucks)



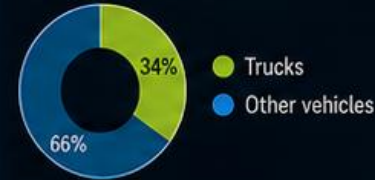
~5,600 e-trucks targeted under PM E-DRIVE

URGENCY

3% of vehicles

34% of road-transport CO₂ emissions

Road-transport CO₂ emissions split



Trucks are the highest-leverage segment to decarbonise.



THE BOTTLENECK

Fleet managers still cannot predict:



Range under real payload



Charging time on the route



Arrival battery and ₹/km



Good economics do not matter if the trip cannot be trusted.



THE CONNECTED-TRUCK RACE HAS ALREADY STARTED

MAHINDRA BLAZO i-TRK

AI chooses diesel driving mode



AI DRIVING MODE
Optimum ON



Fuel Efficiency
+8.6%

Driving Score
92/100



MAHINDRA ZEO + NEMO

Connected EV tracking and trip tools



LIVE VEHICLE
Connected



- Trip Monitor
- Geofencing
- Alerts
- Reports

TATA WHITESPACE

Whole-journey EV intelligence



Route + terrain + weather +
charging + payload



Connectivity is becoming standard. **Accurate trip decisions** can still differentiate.



The e-truck may be economical on paper, but the fleet manager **cannot confidently plan tomorrow's trip**



BROCHURE PROMISE

Certified range, lower running cost



FLEET REALITY

Every route behaves differently

1 · PLAN



FRICTION

Usable range is unclear



CONSEQUENCE

Extra safety buffer added

2 · LOAD



FRICTION

Payload changes consumption



CONSEQUENCE

Fewer orders assigned

3 · DRIVE



FRICTION

Traffic and gradients disrupt the plan



CONSEQUENCE

Delivery window comes under pressure

4 · CHARGE



FRICTION

Stop duration and availability vary



CONSEQUENCE

Productive hours are lost

5 · REVIEW



FRICTION

Actual ₹/km is hard to explain



CONSEQUENCE

Next purchase is delayed

WHAT THE FLEET DOES



Restricts e-trucks to familiar routes



Runs below practical capacity



Keeps diesel as backup



Delays fleet-wide conversion

ROOT CAUSE CHAIN



Range uncertainty



Defensive planning



Lower utilisation



Weaker business case



Slower adoption



The core issue is repeatability: can the truck complete the same route reliably, day after day?

Business challenge only · technology response follows

Move from **selling certified range** to delivering **predictable trip economics**

The strategic prize is not another feature; it is a longer, more valuable fleet relationship.



START WITH ADVICE •
EARN THE RIGHT TO GUARANTEE

1 + OWN THE RANGE PROBLEM

Give every trip a reliable route, speed and charging plan

KPI Range confidence



Route intelligence



Energy modeling



Charging availability



Risk & buffer management

2 • MAKE OPERATIONS PREDICTABLE

Forecast arrival battery, trip time and ₹/km before dispatch

KPI Fleet uptime + cost/km



Trip & ETA forecasting



Dispatch optimization



Exception management



Cost per km control

3 • MONETISE PERFORMANCE

Bundle software, charging and service into cost-per-km contracts



Recurring revenue + EV conversion



Fleet OS platform



Charging network access



Service orchestration



Cost-per-km contracts

THE BUSINESS MODEL SHIFT

TODAY



FUTURE



Vehicle sale



Lifecycle relationship



Certified range



Route-specific confidence



One-time margin



Recurring software revenue



Basic telematics



Decision support



Product warranty



Performance commitment

THE DATA FLYWHEEL



Better trip decisions



More electric trips



Richer operating data



Stronger predictions



More credible guarantees



More e-truck sales



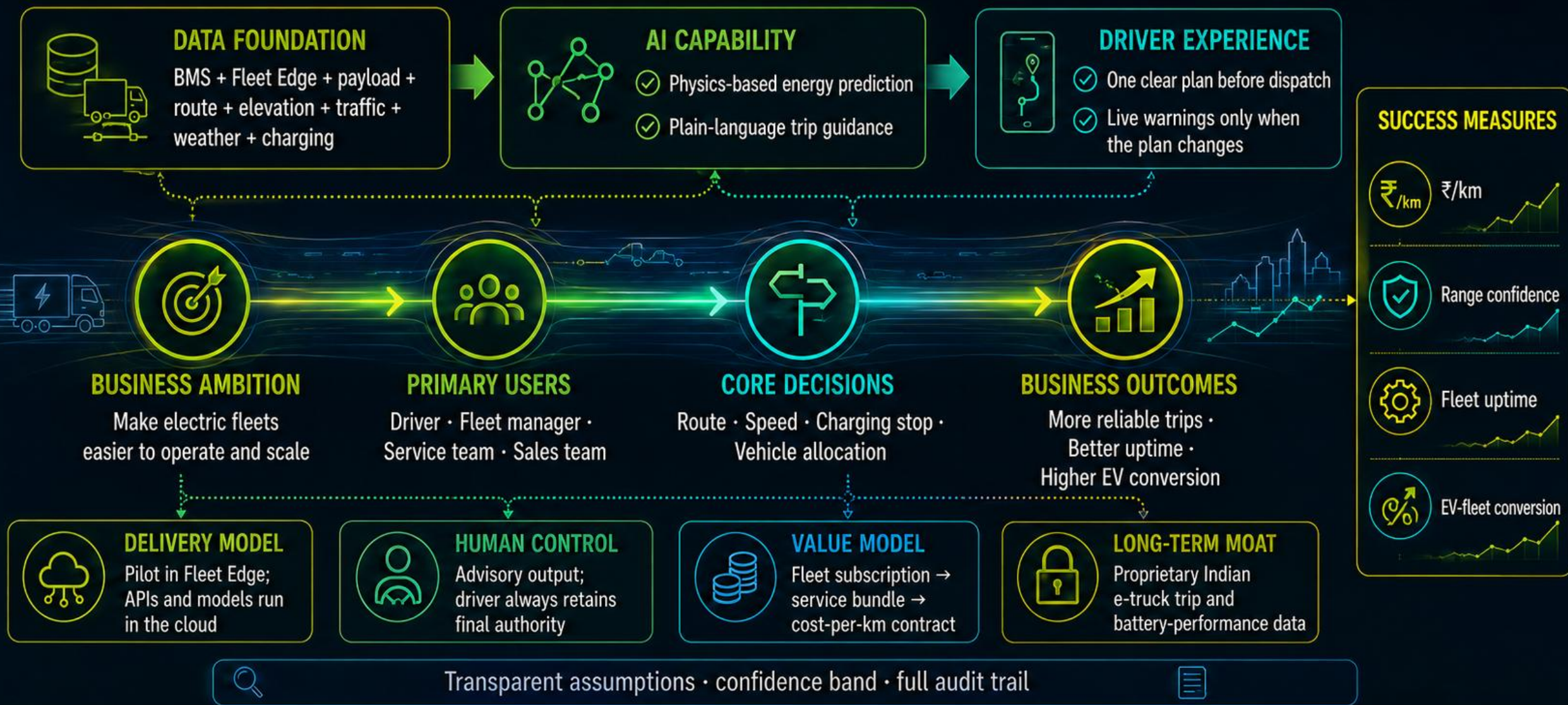
RangeSure becomes the wedge; the connected-fleet platform becomes the moat.

Range optimisation is the right first move: high value, feasible now and useful for every electric trip



Start with the trip. Expand into the fleet. Monetise the platform.

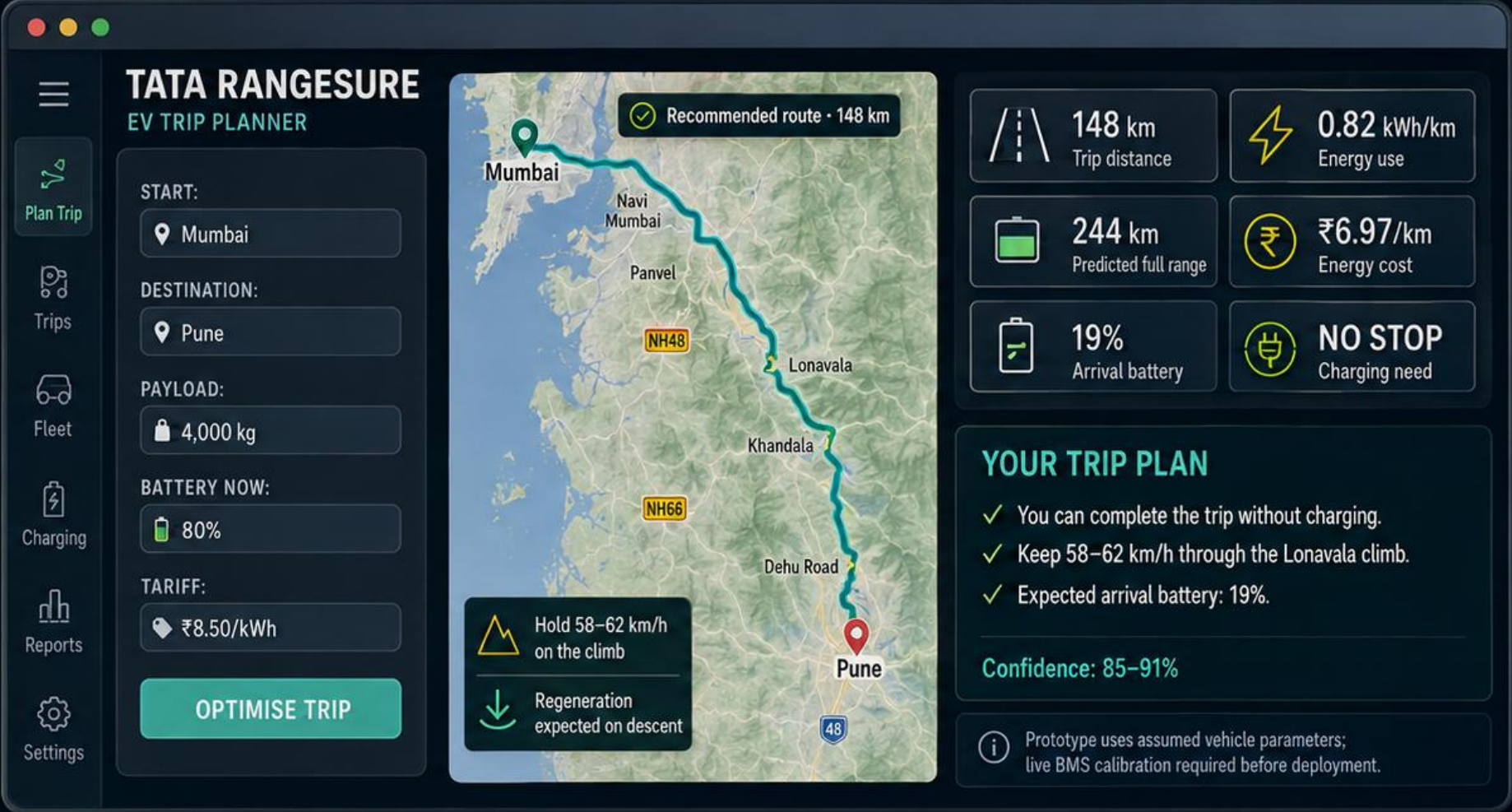
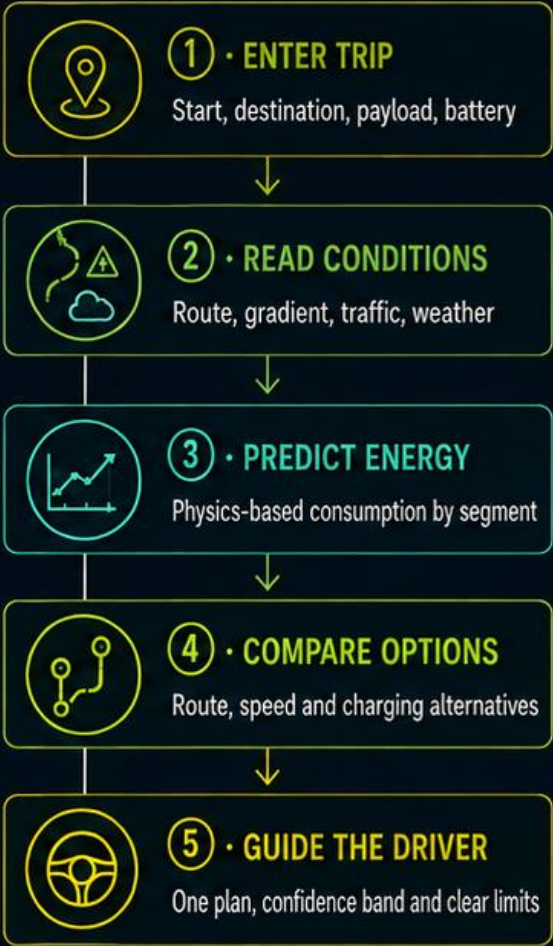
Build RangeSure as the first layer of a broader **connected-fleet intelligence platform**



RangeSure is not another dashboard. It is **the decision layer** connecting **Tata's vehicles, data and fleet relationships.**

RangeSure answers one practical question before dispatch: will this truck complete the trip safely?

FROM INPUT TO DECISION



PREDICTION

Model estimates energy



EXPLANATION

Guidance explains why



DRIVER DECISION

Driver keeps control



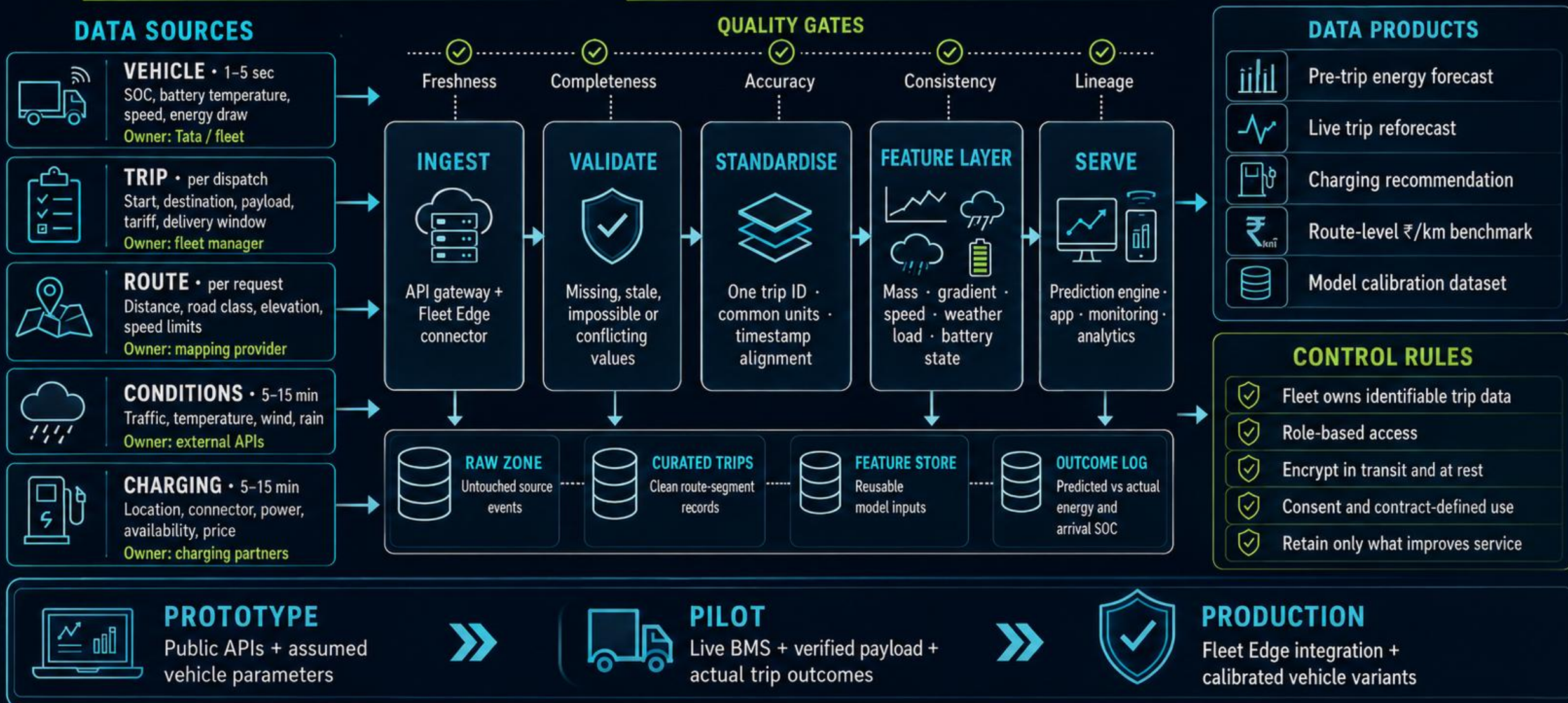
ILLUSTRATIVE PROTOTYPE OUTPUT • replace with live app result

A 100-truck pilot can create ₹1.27 crore of measurable value over three years in the base case

The case works only if RangeSure improves real operating behaviour; every assumption is a pilot gate.



RangeSure needs no training dataset to start; it needs disciplined trip data to improve



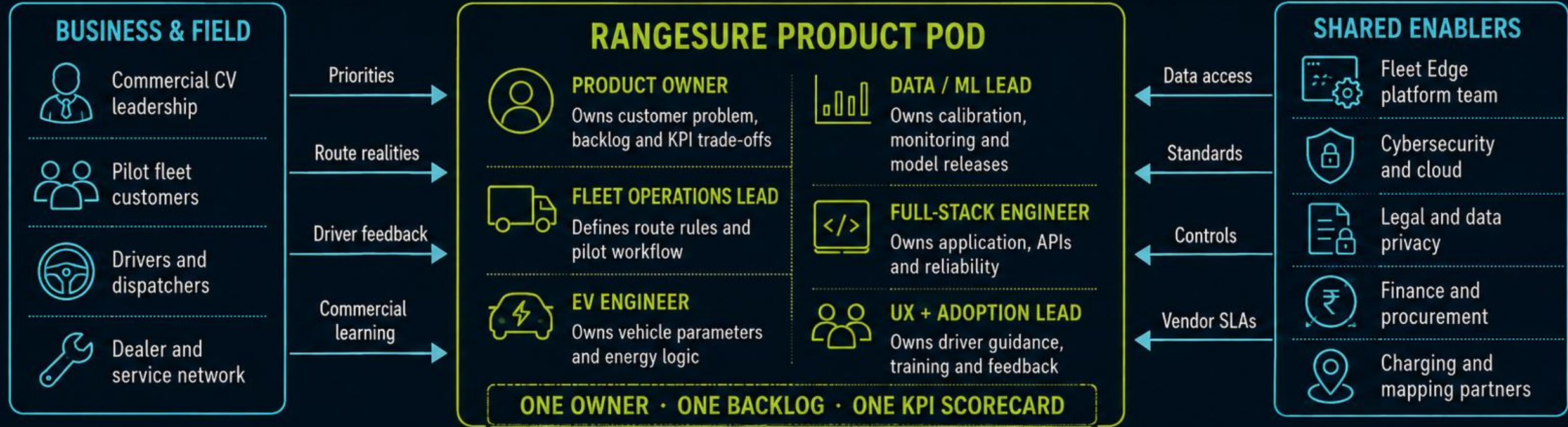
Start transparent. Learn from every trip. Never let convenience weaken data quality.

The prototype is lightweight by design; production adds integration and resilience, not a new concept



 **One architecture, three levels of maturity; no throwaway prototype.** *Reference architecture • final API choices subject to app-team implementation*

RangeSure needs one accountable product pod, not a hand-off between technology and business teams



DECISION RIGHTS		
WHAT CHANGES?		WHO DECIDES?
	Driver guidance copy	Product owner approves
	Physics parameters	EV engineer + ML lead approve
	Safety threshold	Risk owner approves
	Customer rollout	Business sponsor approves
	Emergency rollback	On-call engineer acts immediately



Treat the app as a live fleet product, not a one-time IT project.

Govern RangeSure as high-impact advice: the model recommends, the driver decides

RISK CLASSIFICATION



Operational impact: **HIGH**



Autonomy level: **ADVISORY**



Customer harm if wrong: **DELAY OR STRANDING**



Human override: **ALWAYS AVAILABLE**

DECISION BOUNDARY

AI MAY

- ✓ Estimate energy
- ✓ Compare routes
- ✓ Recommend speed
- ✓ Suggest a charging stop

AI MAY NOT

- ✗ Override the driver
- ✗ Ignore safety reserve
- ✗ Change vehicle controls
- ✗ Hide low confidence

CONTROL STACK



1

• CALCULATION CONTROL

Physics engine owns every numeric output



2

• RULE CONTROL

Hard reserve, charger and data-quality thresholds



3

• EXPLANATION CONTROL

LLM may explain only approved numbers



4

• HUMAN CONTROL

Driver accepts, adjusts or rejects the advice



SAFE DEFAULT: when confidence drops, show uncertainty and fall back to manual planning.

ACCOUNTABILITY



Business sponsor
Owns customer and financial outcome



Product owner
Owns feature and rollout decisions



EV engineer
Owns physical validity



Model-risk owner
Owns thresholds and independent review



Driver / dispatcher
Owns final trip decision

EVIDENCE LOG

- Inputs and data freshness
- Model and rules version
- Prediction and confidence
- Recommendation shown
- Override and reason
- Actual trip outcome

RELEASE GATES

DESIGN



Document assumptions

VALIDATE



Back-test routes

SHADOW MODE



No driver advice

CONTROLLED PILOT



100 trucks · 3 corridors

SCALE



Only after KPI gates

LIVE MONITORING DASHBOARD



Range error



Arrival SOC error



False no-charge advice



API failure rate



Override rate

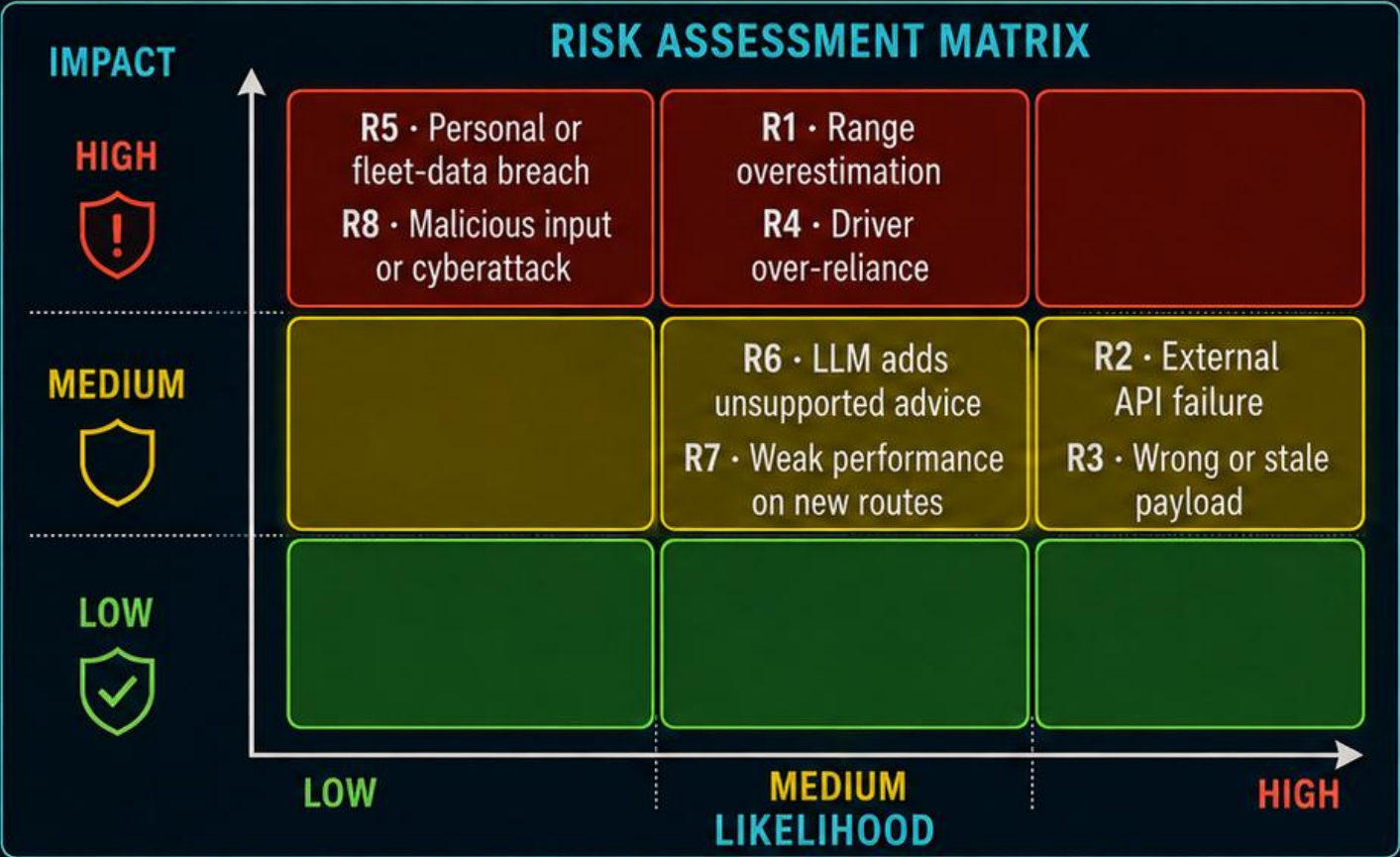


Performance by truck and route



Transparency is the **safety feature**: every recommendation must be explainable, traceable and reversible.

The critical risk is not a wrong number; it is **confident advice that strands a truck**



EARLY-WARNING INDICATORS

False no-charge advice

Arrival SOC error

API latency

Override spike

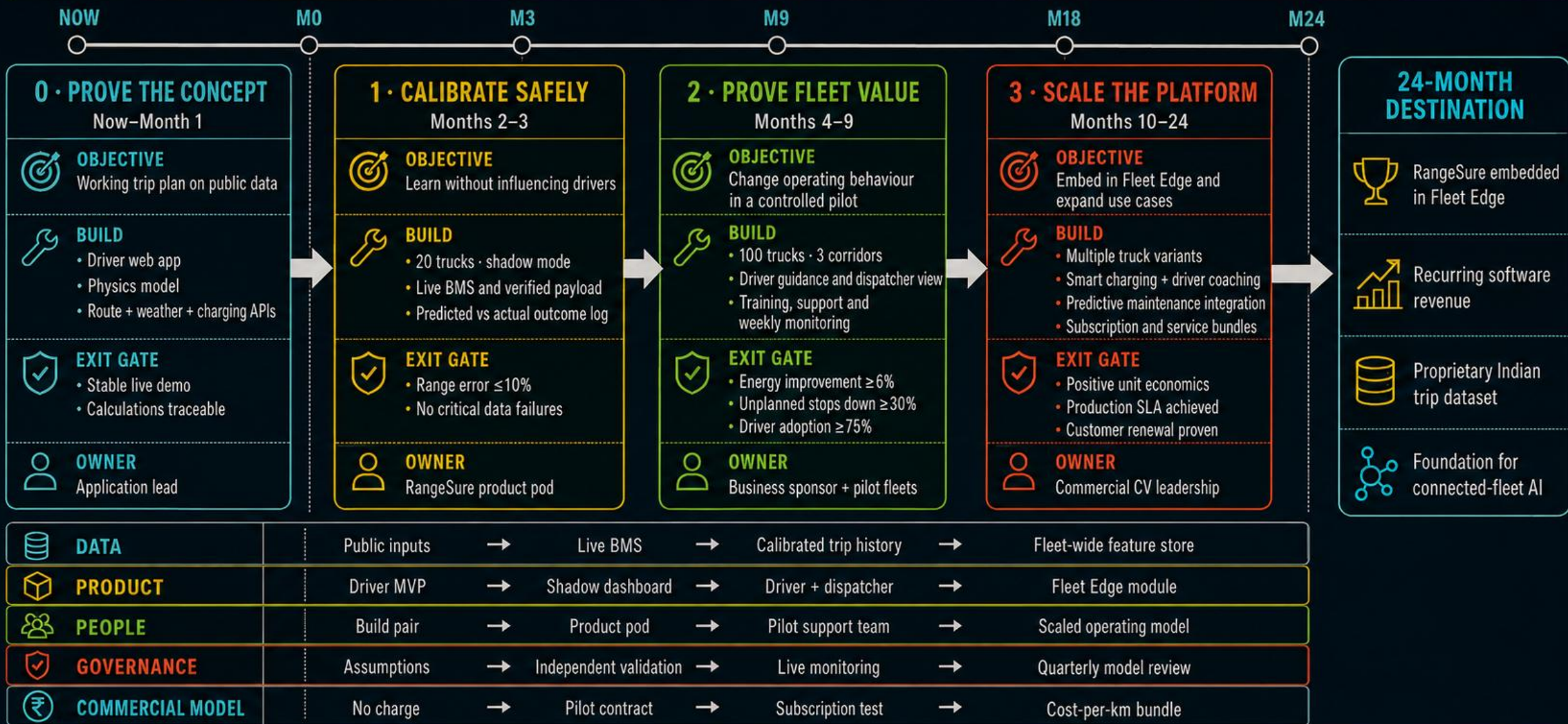
Performance gap by route

MITIGATION REGISTER		
RISK	MITIGATION	OWNER
R1	Reserve threshold + confidence band + shadow testing	Owner: Model risk
R2	Cache, timeout, degraded mode and manual fallback	Owner: Engineering
R3	Payload confirmation and impossible-value checks	Owner: Fleet ops
R4	Advisory wording, training and visible override	Owner: Product
R5	Least access, encryption and retention limits	Owner: Privacy
R6	LLM explains approved fields only; no calculations	Owner: ML lead
R7	Route-level monitoring and controlled expansion	Owner: EV engineer
R8	Input validation, rate limits and security testing	Owner: Cybersecurity

RED LINE: if safety reserve, data freshness or confidence fails, RangeSure **must not** give a go/no-go recommendation.

Residual risk remains; the answer is a safe fallback, not hidden uncertainty.

Scale only after evidence: **prototype** → **shadow mode** → **controlled pilot** → **Fleet Edge product**



Fund the next gate, not the entire promise.

Approve a **controlled pilot** now; commit to scale only after the **economics** and **safety** are proven

DECISION REQUESTED



APPROVE

- ₹35 lakh initial build + integration
- 12-month pilot programme
- 20 trucks in shadow mode
- 100 trucks across 3 corridors after validation



DO NOT APPROVE YET

- Fleet-wide rollout
- Cost-per-km guarantee
- Multi-year platform investment

WHAT THE APPROVAL UNLOCKS



1 · PRODUCT POD

Six cross-functional owners with one backlog



2 · DATA ACCESS

Fleet Edge, BMS, payload and trip outcomes



3 · PILOT CUSTOMERS

Three fleets with repeat, high-utilisation routes



4 · VENDOR ACCESS

Mapping, weather and charging-data services



5 · FIELD ENABLEMENT

Driver training, dispatcher workflow and support



LAUNCH PACKAGE

START SMALL ENOUGH TO CONTROL · LARGE ENOUGH TO LEARN

PASS / PAUSE / SCALE



PASS TO LIVE PILOT

- Range error $\leq 10\%$
- No critical safety failure



PASS TO SCALE

- Energy improvement $\geq 6\%$
- Unplanned stops down $\geq 30\%$
- Driver adoption $\geq 75\%$
- Positive unit economics



PAUSE IMMEDIATELY

- False no-charge advice
- Stale critical data
- Confidence below threshold
- Unresolved privacy or cyber breach

GOVERNANCE CADENCE



Weekly product and safety review



Monthly CFO value review



Quarterly Board checkpoint



Independent validation before scale



BOARD RESOLUTION: Authorise Phase 0–2 of Tata RangeSure within the stated budget and gates; require a new approval before production scale.



This is an approval to *learn with control*, not an approval to overpromise.

The winning e-truck will not claim the longest range; it will make **every kilometre predictable**



1 • SOLVE THE REAL BARRIER

Fleet buyers need confidence in the actual trip, not only the certified range.



2 • USE THE ADVANTAGE TATA ALREADY HAS

Vehicles + Fleet Edge + fleet relationships make RangeSure feasible now.



3 • EARN THE RIGHT TO SCALE

Prove accuracy and savings first; expand into subscriptions and cost-per-km contracts later.



WHY TATA CAN WIN



600,000+ connected vehicles



Growing electric CV portfolio



Live dealer and service network



Trip data that improves with use



TRUSTED TRIP



REPEAT USAGE



BETTER DATA



STRONGER GUARANTEE



MORE EV SALES



APPROVE TATA RANGESURE

Start with a controlled 12-month pilot. Scale only after the four KPI gates are met.



₹/km



Range confidence



Fleet uptime



EV-fleet conversion



One trip decision today. A **connected-fleet platform** tomorrow.

QUESTIONS & BOARD DISCUSSION